Naunyn-Schmiedeberg's Archives of Pharmacology (2024) 397:2551–2566
https://doi.org/10.1007/s00210-023-02781-2

REVIEW

Effects of Chemotherapy on the Immune System: Implications for Cancer Treatment and Patient Outcomes

Anirudh Sharma[1] · Shivam Jasrotia[1] · Ajay Kumar[2]

Received: 14 September 2023 / Accepted: 11 October 2023 / Published online: 31 October 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract

Chemotherapy is a cornerstone of cancer treatment, but it can also induce immune suppression, which can have significant implications for patient outcomes. This review paper aims to give a general overview of how chemotherapy affects the immune system and how it affects cancer treatment. Chemotherapy can directly affect immune cells, leading to cytotoxic effects, cell differentiation and function alterations, and cell communication and signaling pathways disruptions. Such immune suppression can weaken the anti-tumor immune response and increase the risk of immune-related toxicities. Understanding the mechanisms of chemotherapy-induced immune suppression is crucial for optimizing treatment strategies. Strategies to mitigate immune suppression include immunomodulatory agents as adjuvants to chemotherapy, combination therapies to enhance immune function, and supportive care measures of the immune system. Additionally, identifying potential biomarkers to predict immune suppression and guide treatment decisions holds promise for personalized cancer medicine. Future directions in this field involve further elucidating underlying mechanisms, exploring novel combination therapies, and developing targeted interventions to minimize immune suppression. By understanding and addressing chemotherapy-induced immune suppression, we can optimize cancer treatment strategies, enhance the anti-tumor immune response, and improve patient outcomes.

Keywords Anti-tumor · Cancer treatment · Chemotherapy · Cytotoxicity · Immunomodulatory · Immune system · Immune suppression

Introduction

Chemotherapy, a widely used treatment modality for various types of cancer, has proven effective in reducing tumor burden and improving patient outcomes (Galluzzi et al., 2012). However, emerging evidence suggests that chemotherapy can also profoundly affect the immune system, which plays a crucial role in tumor surveillance and eradication (Galluzzi et al., 2012). Understanding the intricate interplay between chemotherapy and the immune system is essential for optimizing cancer treatment strategies and minimizing potential adverse effects. Chemotherapy agents exert their cytotoxic effects primarily by targeting rapidly dividing cells, including cancerous and healthy cells (Weiskopf & Weissman, 2021). While the primary goal of chemotherapy is to eliminate cancer cells, it can inadvertently impact immune cells, leading to immune suppression or dysregulation (Mackall et al., 1995). The immune system consists of a complex network of cells, tissues, and molecules that work together to recognize and eliminate foreign or abnormal cells, including cancer cells (Casares et al., 2005). Any disruption in this delicate balance can have significant implications for both the effectiveness of chemotherapy and the overall immune response (Zitvogel et al., 2008). Several mechanisms contribute to chemotherapy-induced immune suppression (Fig. 1). Direct cytotoxic effects of chemotherapy agents on immune cells, such as T-lymphocytes, natural killer

✉ Shivam Jasrotia
shivamzoology@gmail.com

Anirudh Sharma
sharmaani316@gmail.com

Ajay Kumar
kumar.ajay1250@gmail.com

[1] Department of Biosciences (UIBT), Chandigarh University, Mohali, Punjab 140413, India

[2] University Center for Research & Development (UCRD), Chandigarh University, Gharuan, Mohali, Punjab 140413, India

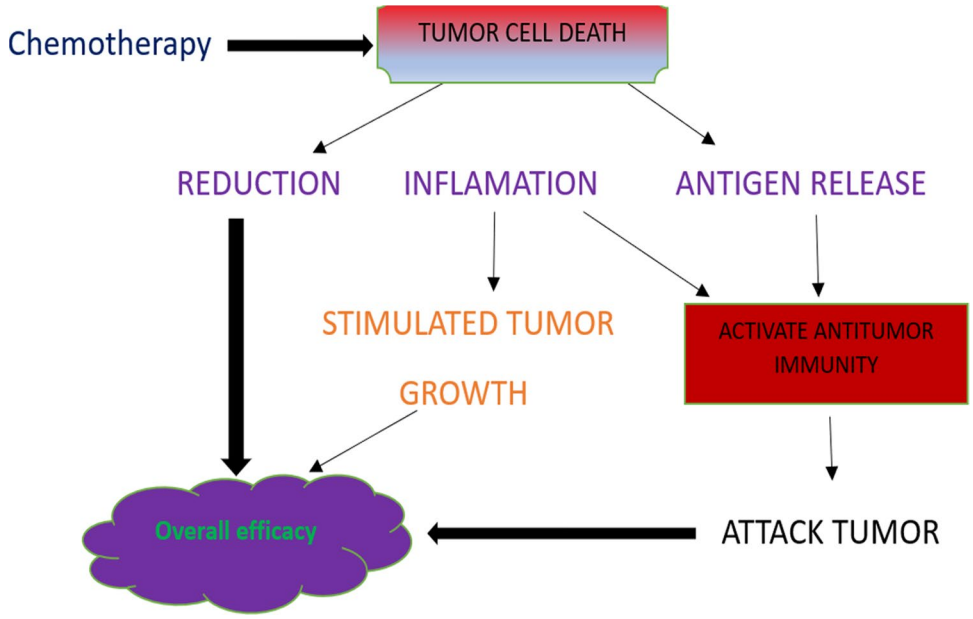

Fig. 1 Oncotarget, tumor reductive therapies and anti-tumor immunity

(NK) cells, and dendritic cells (DCs), can impair their function and reduce their numbers (Galluzzi et al., 2012). Additionally, chemotherapy can alter the differentiation and evolution of immune cells, leading to dysfunctional immune responses.

Furthermore, chemotherapy may disrupt communication and signaling pathways among immune cells, compromising their coordinated efforts to mount an effective anti-tumor response (Sistigu et al., 2014). It is crucial to explore the impact of chemotherapy on innate and adaptive immunity. Innate immune cells, including NK cells, DCs, macrophages, and neutrophils, play a critical role in the initial recognition and elimination of tumor cells. Chemotherapy can modulate the activity and function of these cells, potentially impairing their ability to mount an effective immune response (Weiskopf & Weissman, 2021). Similarly, adaptive immune cells, particularly T-lymphocytes and B-lymphocytes, undergo significant alterations in response to chemotherapy, affecting their proliferation, activation, and effector functions. (Sharma & Allison, 2015).

Immune checkpoint inhibitors (ICIs) have revolutionized cancer treatment in the past few years by using the immune system to target cancer cells specifically. Understanding the interaction between chemotherapy and ICIs is of utmost importance, as combining these therapies may offer synergistic effects or potential challenges due to overlapping immune-related toxicities (Galluzzi et al., 2012). In summary, chemotherapy has wide-ranging effects on the immune system, significantly impacting the efficacy of cancer treatment and patient outcomes. This article aims to comprehensively analyze the mechanisms underlying chemotherapy-induced immune suppression and its implications for cancer therapy. By elucidating the complex relationship between chemotherapy and the immune system, we can develop strategies to minimize immune-related toxicities and enhance the therapeutic potential of chemotherapy (Sharma & Allison, 2015).

The immune system plays a critical role in cancer development, progression, and response to treatment (Fig. 2). It orchestrates a complex network of immune cells, signaling

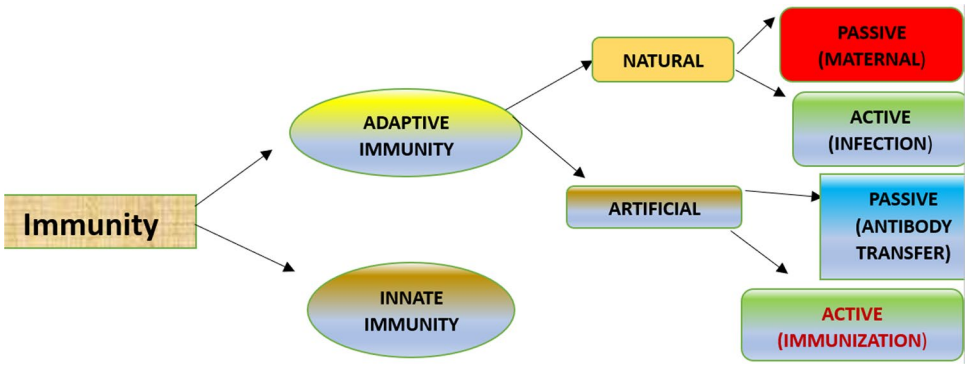

Fig. 2 Flow chart showing the development of immunization

molecules, and checkpoints to recognize and eliminate malignan T-cells. Chemotherapy, a cornerstone of cancer treatment, has traditionally focused on directly targeting and killing cancer cells (Topalian et al., 2015). However, emerging evidence suggests that the effects of chemotherapy extend beyond tumor cell destruction and can significantly impact the immune system. Understanding these effects is of paramount importance for several reasons.

Firstly, a comprehensive understanding of chemotherapeutic effects on the immune system allows for optimizing cancer treatment strategies (Postow et al., 2015). By elucidating the specific immune alterations induced by chemotherapy, clinicians and researchers can identify potential synergistic interactions between chemotherapy and immunotherapies. This knowledge can guide the development of combination therapies that harness the immune system's power to enhance the overall treatment response (Postow et al., 2015). Moreover, understanding the immune-related toxicities associated with chemotherapy enables the implementation of appropriate supportive care measures to minimize adverse effects and improve patient tolerance to treatment.

Secondly, deciphering the intricate interplay between chemotherapy and the immune system can help explain treatment outcomes and patient responses (Wang et al. 2017). It has been observed that the immune status of patients before and during chemotherapy influences treatment efficacy and prognosis. By characterizing the immune profile of patients, including immune cell subsets, cytokine profiles, and immune checkpoint expression, clinicians can identify predictive biomarkers that inform treatment decisions (Postow et al., 2015). This personalized approach may improve patient stratification, more tailored treatment regimens, and enhanced therapeutic outcomes. Thirdly, understanding chemotherapeutic effects on the immune system provides insights into the mechanisms underlying treatment resistance and disease relapse (Ruffell & Coussens, 2015). Chemotherapy-induced immune suppression or dysregulation can create a permissive environment for cancer cells to evade immune surveillance and develop resistance to treatment. Investigating the immune evasion mechanisms employed by cancer cells in response to chemotherapy can shed light on potential therapeutic targets for overcoming treatment resistance (Ruffell & Coussens, 2015). Additionally, unravelling the factors contributing to immune reconstitution post-chemotherapy can aid in developing strategies to restore immune function and prevent disease recurrence. Lastly, exploring the effects of chemotherapy on the immune system can drive the development of novel therapeutic approaches. The emerging field of immunotherapy, including immune checkpoint inhibitors, adoptive cell therapies, and cancer vaccines, has revolutionized cancer treatment by leveraging the immune system's inherent anti-tumor capabilities (Wang et al. 2017). Understanding how chemotherapy modulates immune responses can guide the rational design of combination therapies that capitalize on the strengths of both approaches. Such synergistic combinations promise to enhance treatment responses, extend survival rates, and potentially achieve long-term remission in cancer patients.

Mechanisms of Chemotherapy-Induced Immune Suppression

The mechanisms underlying chemotherapy-induced immune suppression are multifaceted and involve direct cytotoxic effects on immune cells, alterations in immune cell differentiation and function, and disruption of immune cell communication and signaling pathways. Here's a sample elaboration on each subheading.

Direct Cytotoxic Effects on Immune Cells

Chemotherapy agents, primarily targeting rapidly dividing cancer cells, can also exert direct cytotoxic effects on various immune cell populations (Table 1) (Vincent & Mignot, 2018). Chemotherapy can impact immune cells such as T-lymphocytes, natural killer cells (NK cells), and dendritic cells (DCs), decreasing immunological function. For example, some chemotherapeutic agents may induce apoptosis or inhibit the proliferation of T-lymphocytes, which are essential for orchestrating immune responses against cancer cells (Gajewski et al. 2006). Similarly, NK cells, which play a crucial role in recognizing and eliminating tumor cells, can be susceptible to chemotherapy-induced cell death or functional impairment (Ramakrishnan et al., 2020). DCs, critical for initiating and coordinating immune responses, may also experience reduced antigen-presenting capacity or impaired maturation following exposure to chemotherapy agents (Del Prete et al. 2023). These direct cytotoxic effects on immune cells contribute to immune suppression and compromise the overall anti-tumor immune response (Ramakrishnan et al., 2020).

Alterations in Immune Cell Differentiation and Function

Chemotherapy can disrupt the delicate balance of immune cell differentiation and function. Immune cells undergo complex processes of maturation and differentiation to acquire specialized functions necessary for mounting effective immune responses (Galluzzi et al., 2015). Chemotherapy agents can interfere with these processes, altering immune cell populations and their functional properties. For instance, certain chemotherapeutic drugs may inhibit

Table 1 Impact of Chemotherapy on Innate Immunity

Impact of Chemotherapy on Innate Immunity	Explanation	Reference
Neutropenia	Chemotherapy can cause a decrease in the number of neutrophils, which are key immune cells involved in innate immunity. This increases the risk of infections. Chemotherapy, a cornerstone of cancer treatment, exerts profound effects on various aspects of the immune system's innate defences. One crucial impact is the development of neutropenia, a decrease in neutrophil counts, the body's primary defence against infections. This decline heightens the susceptibility to various pathogens, increasing the risk of potentially severe infections. Additionally, chemotherapy's influence extends to macrophages, integral components of innate immunity that engulf and neutralize pathogens. Chemotherapy drugs can impair their functionality, compromising the body's capacity to fend off invaders effectively.	(Koh et al., 2009)
Macrophage Dysfunction	Chemotherapy drugs may affect the function of macrophages, impairing their ability to engulf and destroy pathogens effectively.	(Shapouri-Moghaddam et al., 2018)
Natural Killer (NK) Cell Suppression	Chemotherapy can suppress the activity of NK cells, which play a crucial role in recognizing and killing cancer cells and infected cells. Another consequence involves the suppression of Natural Killer (NK) cells, which play a pivotal role in recognizing and eliminating both cancer cells and infected cells. Chemotherapy-induced NK cell suppression can diminish their anti-tumor and anti-viral activities, weakening the immune response. Toll-like receptors (TLRs), essential in pathogen recognition, can also be affected, leading to their downregulation. As a result, the innate immune response's ability to detect and respond promptly to pathogens might be compromised.	(Cao et al., 2020)
Toll-like Receptor (TLR) Downregulation	Chemotherapy may downregulate the expression of Toll-like receptors, reducing the innate immune response to pathogen recognition.	(Janssens & Beyaert, 2003)
Reduced Cytokine Production	Chemotherapy can lead to a decrease in the production of pro-inflammatory cytokines, which are important for initiating and regulating immune responses.	(Seruga et al., 2008)
Epithelial Barrier Dysfunction	Chemotherapy can disrupt the integrity of the epithelial barrier, making it easier for pathogens to enter the body and cause infections. Moreover, chemotherapy's influence extends to structural barriers. The treatment can disrupt the epithelial barrier's integrity, which serves as a frontline defence against invading pathogens. This disruption facilitates the entry of microbes and heightens the risk of infections. The complement system, responsible for opsonization and lysis of pathogens, can also be impaired by chemotherapy drugs, potentially weakening the immune system's ability to neutralize pathogens.	(Teoh & Pavelka, 2016)
Altered Complement System Function	Chemotherapy drugs can affect the activity of the complement system, impairing its ability to eliminate pathogens through opsonization and lysis.	(Merle et al., 2015)
Changes in Dendritic Cell Function	Chemotherapy may alter the function of dendritic cells, which are responsible for initiating adaptive immune responses by presenting antigens to T-cells. Chemotherapy's intricate impact on immune cells includes altering dendritic cell function. Dendritic cells are key players in initiating adaptive immune responses, acting as messengers between innate and adaptive immunity. Disruption of their function by chemotherapy could impede the body's ability to mount effective immune responses against diverse threats.	(Constantino et al., 2017)

Table 1 (continued)

Impact of Chemotherapy on Innate Immunity	Explanation	Reference
MasT-cell Activation	Chemotherapy drugs can stimulate masT-cell activation, leading to the release of inflammatory mediators and allergic-type reactions. Furthermore, chemotherapy-induced mast T-cell activation can lead to the release of inflammatory mediators, potentially resulting in allergic reactions or inflammation. Lastly, impaired wound healing due to chemotherapy's effects on immune cells involved in tissue repair can contribute to delayed recovery and complications after surgery or injuries.	(Beghdadi et al., 2011)
Impaired Wound Healing	Chemotherapy may delay the wound healing process by affecting the migration and function of immune cells involved in tissue repair.	(Deptuła et al., 2019)

the differentiation of precursor cells into mature immune cells, resulting in reduced numbers or functional deficiencies of specific immune cell subsets (Galluzzi et al., 2015). Moreover, chemotherapy can perturb the cytokine and chemokine microenvironment, influencing immune cell polarization and functional phenotypes. These alterations in immune cell differentiation and function can impair immune surveillance and compromise the ability to mount an effective anti-tumor immune response (Garg & Agostinis, 2017).

Disruption of Immune Cell Communication and Signaling Pathways

Effective immune responses rely on precise communication and signaling among different immune cell populations (Reits et al., 2006). Chemotherapy can disrupt this communication network by interfering with immune cell signaling pathways. For instance, certain chemotherapy agents may impact the production or responsiveness to key cytokines and chemokines involved in immune cell activation and migration. Additionally, chemotherapy can modulate the expression and function of immune checkpoint molecules, which regulate immune cell activation and tolerance (Zou et al., 2016). Disruption of these communication and signaling pathways can lead to dysregulated immune responses, immune cell exhaustion, and impaired anti-tumor immunity.

Overall, the mechanisms of chemotherapy-induced immune suppression involve direct cytotoxic effects on immune cells, alterations in immune cell differentiation and function, and disruption of immune cell communication and signaling pathways. Understanding these mechanisms is crucial for developing strategies to mitigate immune suppression, enhance the efficacy of chemotherapy, and design novel therapeutic approaches that integrate immunomodulatory strategies to optimize cancer treatment outcomes.

Effects of Chemotherapy on Adaptive Immunity

Impact on T-Lymphocytes

T-lymphocytes play a crucial role in adaptive immune responses, including recognizing and eliminating cancer cells. Chemotherapy can profoundly affect T-lymphocytes, influencing their proliferation, activation, and effector functions (Rabinowich et al., 1994). Here are some key impacts of chemotherapy on T-lymphocytes:

Reduction in T-lymphocyte numbers

Certain chemotherapeutic agents i.e. Monoclonal antibodies, immunosuppressants, radiation therapy, inhibitors, cytotoxic drugs, antimetabolites can cause a decrease in the total count of T-lymphocytes, a condition known as lymphopenia (Schirrmacher, 2019). This reduction in T-lymphocyte numbers can impair immune surveillance and reduce capacity for mounting an effective immune response against cancer cells (Vincent & Mignot, 2018). Lymphopenia resulting from chemotherapy can weaken the immune system's ability to mount an effective immune response against cancer cells (Lee et al., 2018). This is because T-lymphocytes are a key component of the adaptive immune system, specifically targeting and eliminating cancerous cells. When their numbers are reduced, the immune surveillance is compromised, and the body's ability to recognize and destroy cancer cells is hindered Lee et al., 2018). To mitigate the impact of lymphopenia, oncologists and healthcare providers often monitor the patient's immune status during chemotherapy treatment. In some cases, they may adjust the chemotherapy regimen or prescribe medications to support the immune system, such as growth factors that stimulate the production of immune cells (Gao et al., 2020a, b).

Impaired T-cell proliferation

Chemotherapy agents targeting rapidly dividing cells can affect cancer cells and rapidly dividing immune cells, including T-lymphocytes. Consequently, the proliferation and expansion of T-cells in response to antigenic stimulation may be compromised, leading to reduced T-cell responses against tumor-associated antigens (Rabinowich et al., 1994). T-cell proliferation is crucial in mounting an effective immune response against cancer cells. When T-cells encounter antigens derived from cancer cells, they undergo activation and start to proliferate (Ma et al., 2013). This proliferation expands the specific T-cell population capable of recognizing and attacking the cancer cells. However, chemotherapy drugs that affect rapidly dividing cells may also target dividing T-cells. This can lead to a decrease in T-cell proliferation and compromise the expansion of tumor-specific T-cell populations. As a result, the immune response against tumor-associated antigens may be dampened or insufficient (Klapper et al., 2008).

Disrupted T-cell signaling and activation

Chemotherapy can interfere with the signaling pathways required for T-cell activation. This disruption can affect the recognition of antigen-presenting cells, co-stimulatory signals, and the formation of the immunological synapse (Seo & Jeong, 2020). Consequently, impaired T-cell activation hampers the development of effective anti-tumor immune responses. Chemotherapy agents can interfere with these signaling pathways, affecting the activation and function of T-cells. This disruption can occur through various mechanisms, such as direct toxicity to T-cells or interference with the signaling molecules involved in T-cell activation. A series of signaling events are initiated when T-cells encounter antigens presented by antigen-presenting cells (APCs), such as dendritic cells. This leads to the activation of T-cell receptors (TCRs) and the subsequent release of cytokines and other immune molecules that contribute to an effective immune response. Chemotherapy drugs can interfere with these signaling events, impairing T-cell activation and reduced immune responses against cancer cells. This disruption can affect the production of cytokines, the expression of co-stimulatory molecules, and the overall T-cell activation process. As a result, the ability of T-cells to effectively recognize and eliminate cancer cells may be compromised (O'Donnell et al., 2019).

Altered T-cell function

Chemotherapy can affect the functional properties of T-lymphocytes. For example, some chemotherapeutic agents may impair the production of cytokines by T-cells, such as interferon-gamma (IFN-γ) and interleukin-2 (IL-2), which are crucial for immune cell communication and effector functions (Choudhury et al., 2018). Additionally, the cytotoxic activity of T-cells may be compromised, leading to reduced tumor cell killing capacity.

Chemotherapy agents can affect T-cell function through various mechanisms. Some chemotherapy drugs can directly damage T-cells or interfere with their function (Seo & Jeong, 2020). Others may indirectly affect T-cell function by targeting other immune system components or causing systemic effects that impact immune responses. Understanding the impact of chemotherapy on T-lymphocytes is vital for optimizing cancer treatment strategies (Hurez et al., 2015). Strategies that aim to mitigate these effects on T-cells, such as using immune-modulating agents or adoptive T-cell therapies, might improve patient outcomes by enhancing the anti-tumor immune response.

Influence on B-Lymphocytes and Antibody Production: -

B-lymphocytes produce antibodies that recognise and neutralise particular antigens, a crucial part of adaptive immunity, including those expressed by cancer cells. Chemotherapy can affect B-lymphocytes, influencing their development, function, and antibody production (De Milito & Younes, 2008). Here are some key influences of chemotherapy on B-lymphocytes and antibody production:

Suppression of B-lymphocyte maturation

Chemotherapy agents, particularly those that target rapidly dividing cells, can interfere with B-lymphocyte maturation (Lenz et al., 2015). This disruption can decrease the number of mature B-cells available for antibody production, affecting the overall immune response against cancer cells. During maturation, B-cells undergo various stages of development in the bone marrow and lymphoid tissues, leading to their differentiation into mature B-cells capable of producing specific antibodies (levi et al., 2018). However, chemotherapy drugs can disrupt this maturation process and impede the production of mature B-cells.

Chemotherapy agents that target rapidly dividing cells can affect cancer cells and normal cells that undergo rapid division, including developing B-cells (Ujjani et al., 2020). The interference with B-cell maturation can decrease the number of mature B-cells available for antibody production.

Impaired antibody production

Chemotherapy-induced immune suppression can hinder the ability of B-lymphocytes to produce antibodies.

Chemotherapeutic agents may directly affect B-cells, leading to decreased antibody production or altering the quality and specificity of the antibodies generated (De Milito & Younes, 2008). This impairment in antibody production can compromise the humoral immune response against cancer cells and decrease the effectiveness of immunotherapy strategies that rely on antibody-mediated mechanisms. The impairment in antibody production caused by chemotherapy can be temporary or long-lasting, depending on various factors such as the specific chemotherapy agents used, dosage, treatment duration, and individual patient factors (Poulain et al., 2019). Healthcare providers need to monitor the immune status, including antibody production, during and after chemotherapy to assess the impact on the humoral immune response. In some cases, additional interventions may be considered to support antibody production, such as immunomodulatory therapies or growth factors that stimulate B-cell recovery (Ansell et al., 2015). These approaches aim to restore and enhance the humoral immune response against cancer cells.

Alteration of antibody isotype distribution

Chemotherapy can influence the production of different antibody isotypes by B-lymphocytes. For example, some chemotherapeutic agents may preferentially affect the production of specific antibody isotypes, such as immunoglobulin M (IgM) or immunoglobulin G (IgG). This alteration in antibody isotype distribution can impact immune responses and the effectiveness of antibody-mediated immune mechanisms (Nimmerjahn & Ravetch, 2008). Antibodies, also known as immunoglobulins (Igs), come in different isotypes, such as IgM, IgG, IgA, IgE, and IgD. Each isotype has unique properties and functions in the immune system (Casadevall & Dadachova, 2004). They play crucial roles in recognizing and eliminating pathogens, including cancer cells. Chemotherapy agents can selectively impact B-cells' production of specific antibody isotypes. Some chemotherapeutic drugs may preferentially affect the production of certain antibody isotypes, such as IgM or IgG (Pone & Zan, 2012). This alteration in antibody isotype distribution can have several consequences

Disruption of B-cell-antigen presentation interactions

Chemotherapy agents can affect the interaction between B-lymphocytes and antigen-presenting cells (APCs), such as dendritic cells (Abu-Eid et al., 2017). This disruption may impair the efficient presentation of tumor-associated antigens to B-cells, hindering the generation of antigen-specific B-cell responses and subsequent antibody production. Chemotherapy drugs can interfere with this interaction between B-cells and APCs in several ways. They may directly affect the function or viability of APCs, leading to impaired antigen presentation. Alternatively, chemotherapy drugs may indirectly impact the interaction by altering the tumor microenvironment or affecting immune cell trafficking.

Impact on memory B-cells

Chemotherapy may also influence the generation and maintenance of memory B-cells, which play a vital role in long-term immune protection and immunological memory. Chemotherapeutic agents can affect the survival and function of memory B-cells, potentially compromising the ability to mount effective recall responses upon re-exposure to cancer cells or specific antigens (Shalapour & Karin, 2015). The impact of chemotherapy on memory B-cells can vary depending on the specific drugs used, their dosage, treatment protocols, and individual patient factors. Monitoring the immune status, including the presence and functionality of memory B-cells, is essential during chemotherapy treatment (Crotty 2019). In some cases, strategies may be employed to support the generation and maintenance of memory B-cells, such as targeted immunomodulatory therapies or the use of adjuvants (Bende et al., 2011). These interventions aim to enhance the development of memory B-cells and improve long-term immune protection against cancer cells or specific antigens.

Effects on Tumor-Infiltrating Lymphocytes (TILs)

Tumor-infiltrating lymphocytes (TILs) are immune cells that have migrated into the tumor microenvironment and play a crucial role in the anti-tumor immune response. Chemotherapy can directly and indirectly affect TILs, influencing their composition, activation, and functional properties (Facciabene & Motz, 2012). Here are some key effects of chemotherapy on TILs:

Alterations in TIL composition

Chemotherapy can impact the composition and distribution of TIL subsets within the tumor microenvironment. Depending on the specific chemotherapy regimen, there may be changes in the relative abundance of T-cell subsets, such as CD4+ helper T-cells, CD8+ cytotoxic T-cells, and regulatory T-cells (Tregs). These alterations can influence the balance between pro-tumor and anti-tumor immune responses within the tumor microenvironment (Gao et al., 2020). The modifications in TIL composition caused by chemotherapy can have implications for the overall immune response within the tumor microenvironment.

Changes in the balance between pro-tumor and anti-tumor immune responses may impact the effectiveness of immunotherapy or other immune-based treatments targeting the tumor (Tumeh et al., 2014). Monitoring the immune status within the tumor microenvironment during chemotherapy treatment is important to assess the impact on TIL composition and the potential effects on treatment outcomes. It can help guide treatment decisions and develop personalized therapeutic strategies, such as combining chemotherapy with immunomodulatory agents to enhance anti-tumor immune responses (Gajewski et al., 2013).

Reduction in TIL numbers

Some chemotherapy agents, particularly those with cytotoxic properties, can cause a reduction in the overall number of TILs within the tumor microenvironment (Topalian et al ., 2015). This decrease in TIL numbers may impair the immune surveillance and anti-tumor activity within the tumor, potentially compromising the effectiveness of the immune response against cancer cells (Fridman et al., 2012) ...Chemotherapy drugs that have cytotoxic effects on rapidly dividing cells, including cancer cells, can inadvertently affect TILs as well. The reduction in TIL numbers can occur due to direct toxicity to T-cells or the impact on the tumor microenvironment. This leads to T-cell depletion or impaired migration into the tumor.

Impaired TIL activation and function

Chemotherapy can affect the activation and functional properties of TILs. Certain chemotherapeutic agents may inhibit T-cell activation signals or disrupt the communication between TILs and other immune cells, leading to impaired effector functions and reduced anti-tumor immune responses (Gao et al., 2020). Additionally, chemotherapy-induced immune suppression can contribute to TIL exhaustion, characterized by reduced cytotoxicity and cytokine production. The impaired activation and function of TILs caused by chemotherapy can result in decreased cytotoxicity, reduced cytokine production, and compromised immune surveillance within the tumor microenvironment (Restifo et al. 2012). This can allow tumor cells to evade immune detection and escape immune-mediated elimination. Monitoring the activation and functional status of TILs during chemotherapy treatment is crucial to assess the impact on the anti-tumor immune response (Zou et al., 2016). Healthcare providers may consider strategies to support TIL activation and function, such as immunomodulatory therapies or combination treatments that enhance T-cell responses.

Impact on TIL infiltration and distribution

Chemotherapy can modulate the migration and infiltration of TILs into the tumor microenvironment. Some chemotherapeutic agents may alter the expression of chemokines and adhesion molecules within the tumor, affecting TIL recruitment and distribution (Spranger & Gajewski, 2018). Changes in TIL infiltration patterns can impact the effectiveness of the immune response against cancer cells. Monitoring TIL infiltration and distribution patterns during chemotherapy treatment is important to assess the impact on the anti-tumor immune response (Chao et al., 2010). Strategies to enhance TIL infiltration, such as combining chemotherapy with immunomodulatory agents or other approaches that modify the tumor microenvironment, may be considered to optimize the immune response against cancer cells

Interplay with immunotherapies

Understanding the effects of chemotherapy on TILs is particularly important when considering combination therapies with immunotherapies such as immune checkpoint inhibitors (ICIs) (Shin & Ribas, 2015). Chemotherapy can have synergistic and antagonistic interactions with immunotherapies, and the effects on TILs must be considered for optimal treatment outcomes.

By elucidating the effects of chemotherapy on TILs, researchers and clinicians can develop strategies to enhance the anti-tumor immune response within the tumor microenvironment (Sharma & Allison, 2015). This knowledge can guide the rational design of combination therapies that incorporate chemotherapy and immunotherapies to maximize therapeutic efficacy and improve patient outcomes.

Chemotherapy and Immune Checkpoint Inhibitors (ICIs)

Synergistic Effects of Chemotherapy and ICIs

Combining chemotherapy and immune checkpoint inhibitors (ICIs) has demonstrated promising synergistic effects in cancer treatment (Table 2) (Topalian et al., 2012). Chemotherapy, with its cytotoxic effects on rapidly dividing cells, including cancer cells, can induce immunogenic cell death and release tumor-associated antigens (Topalian et al., 2012). This process enhances the presentation of antigens to immune cells, promoting an anti-tumor immune response (Topalian et al., 2012). However, chemotherapy can also lead to immune suppression, compromising the immune system's ability to effectively respond to cancer cells (Wei et al., 2018). Immune checkpoint inhibitors target inhibitory checkpoints such as PD-1/PD-L1 or CTLA-4, releasing

Table 2 Comparing Chemotherapy and Immune Checkpoint Inhibitors (ICIs) in Cancer Treatment

Aspect	Chemotherapy	Immune Checkpoint Inhibitors (ICIs)	Reference
Mechanism of Action	Indirectly kills rapidly dividing cells, including cancer cells	Blocks inhibitory signals that suppress the immune response against cancer cells	(Lee et al., 2018)
TargeT-cells	Targets rapidly dividing cells, including cancer cells	Targets immune cells, specifically T-cells	(Lee et al., 2016)
Mode of Administration	Can be administered orally, intravenously, or topically	Usually administered intravenously	(Katz et al., 2001)
Overall Response Rate	Varies depending on the type of cancer and patient factors	Varies depending on the type of cancer and patient factors	(Caro et al., 2001)
Treatment Duration	Generally administered in cycles over a defined period	Administered continuously over a longer duration	(Lana et al., 2007)
Side Effects	Can cause various side effects, including hair loss, nausea, and fatigue	Can cause immune-related side effects, such as inflammation of organs or tissues	(Kroschinsky et al., 2017)
Patient Selection	Often used as a first-line or adjuvant treatment for a wide range of cancers	Primarily used in patients with certain types of cancers and specific biomarkers	(Joshi & Badgwell, 2021)

the brakes on immune cells and enhancing their activation and anti-tumor activity (Wei et al., 2018). By combining chemotherapy with ICIs, the cytotoxic effects of chemotherapy can be coupled with the enhanced immune activation and effector functions mediated by ICIs (Wei et al., 2018). This synergistic approach allows for a more comprehensive and robust anti-tumor immune response, potentially leading to improved treatment outcomes, including enhanced tumor regression, prolonged progression-free survival, and improved overall survival rates.

Additionally, combining chemotherapy and ICIs offers the possibility of overcoming treatment resistance and expanding the therapeutic benefits to a broader range of cancer types (Emens, 2018). Ongoing research and clinical trials continue to explore and optimize the use of this combination approach, aiming to harness the full potential of both chemotherapy and ICIs in cancer treatment (Emens, 2018). By understanding the mechanisms underlying the synergy between chemotherapy and ICIs, clinicians and researchers can develop more effective and personalized treatment strategies for cancer patients.

Modulation of Tumor Microenvironment by Chemotherapy

Chemotherapy directly affects cancer cells and significantly modulates the tumor microenvironment, encompassing various non-cancerous cells, extracellular matrix components, and signaling molecules (Shalapour & Karin, 2015). Chemotherapeutic agents can influence the tumor microenvironment through multiple mechanisms (Shalapour & Karin, 2015). Firstly, chemotherapy can induce immunogenic cell death of cancer cells, leading to the release of danger signals and the activation of the innate immune system (Galluzzi et al., 2015). This activation can trigger the recruitment and infiltration of immune cells, such as dendritic cells, natural killer cells, and T-lymphocytes, into the tumor microenvironment (Galluzzi et al., 2015). Additionally, chemotherapy agents can affect the production and secretion of cytokines and chemokines within the tumor, influencing immune cell recruitment, activation, and polarization (Galluzzi et al., 2015). Moreover, chemotherapy can modulate the composition and activity of stromal cells within the tumor microenvironment, including cancer-associated fibroblasts and tumor-associated macrophages (Shalapour & Karin, 2015). These stromal cells play vital roles in tumor progression, angiogenesis, and immune regulation (Shalapour & Karin, 2015). Chemotherapy can alter their function and phenotype, impacting the tumor microenvironment's immunosuppressive characteristics (Shalapour & Karin, 2015). Furthermore, chemotherapy-induced tumor cell death can lead to the exposure of neoantigens and tumor-associated antigens, promoting the priming and activation of adaptive immune responses within the tumor microenvironment (Pitt et al., 2016). This immune activation within the tumor microenvironment can enhance the anti-tumor immune response and contribute to tumor control. Overall, chemotherapy can profoundly affect the tumor microenvironment, transforming it into a more immunogenic and favorable milieu for anti-tumor immune responses (Pitt et al., 2016). This modulation of the tumor microenvironment by chemotherapy highlights its potential synergy with immunotherapies, such as immune checkpoint inhibitors, which can further enhance immune cell activity and anti-tumor effects. In a very brief summary, chemo-immunotherapy appeared to cause higher rates of remission than chemotherapy alone, but this was not always, only infrequently, converted into a significantly higher

rate of cure (Joseph and Joseph 2006). At The University of Texas M.D. Anderson Hospital in Houston, Texas, supervised clinical trials were started in the early 1970s to treat patients with metastatic melanoma and sarcoma using a combination of viral oncolysate-based vaccinations and modern chemotherapy. Based on the commonly held notion that the efficacy of this treatment technique was questionable at the time, it was usually thought to be unsuitable. Cancer vaccinations would not work if you were using an immunosuppressive chemotherapy's side effects (Joseph and Joseph 2006). Over the past 100 years, various forms of cancer vaccines have been developed and used with varying degrees of effectiveness, but notably there have been numerous reports of complete clinical responses (CR) with regression of all detectable illness in patients with advanced malignancies of various sorts by immunomodulation. These CRs are extremely important occurrences because they frequently support long-term sustained survival times of decades or longer, which equates to a clinical cancer cure. The issue has been the often modest and unpredictably high rates of partial response (PR), stable disease (SD), or even CR induction (Coventry 2019). Still, the simple fact that these kinds of clinically helpful responses take place is crucial because it provides strong proof that cancer vaccines represent a potent treatment option that needs more thorough research and development (Coventry 2019).

Strategies to Mitigate Chemotherapy-Induced Immune Suppression

Immunomodulatory Agents as Adjuvants to Chemotherapy

Immunomodulatory agents have shown promise as adjuvants to chemotherapy, aiming to mitigate immune suppression and enhance anti-tumor immune responses (Mokhtari et al. 2017). These agents can modulate immune cell function, promote immune activation, and overcome immune inhibitory signals (Mokhtari et al. 2017). For example, immune checkpoint inhibitors (ICIs) targeting PD-1/PD-L1 or CTLA-4 have been combined with chemotherapy to unleash the full potential of the immune system (Schatton et al. 2010). By blocking inhibitory checkpoints, ICIs can restore T-cell activity and improve anti-tumor responses (Schatton et al. 2010). Other immunomodulatory approaches, such as toll-like receptor agonists, cytokines, or immune stimulants, can also enhance immune cell activation and antigen presentation (Mokhtari et al. 2017). The rational combination of immunomodulatory agents with chemotherapy holds great promise for synergistically boosting the anti-tumor immune response (Mokhtari et al. 2017).

Combination Therapies to Enhance Immune Function

Combination therapies involving chemotherapy and other immune-enhancing approaches can counteract chemotherapy-induced immune suppression (Mokhtari et al. 2017). For instance, cancer vaccines or adoptive cell therapies, such as CAR-T-cell therapy, can be combined with chemotherapy to enhance immune cell recognition and targeting of cancer cells (Mokhtari et al. 2017). These therapies can prime or reprogram immune cells to specifically recognize and attack tumor cells, overcoming immune evasion mechanisms (Mokhtari et al. 2017). Additionally, strategies such as metronomic chemotherapy can promote anti-angiogenic effects and modify the tumor microenvironment to favor immune cell infiltration and activity (Mokhtari et al. 2017). Combining chemotherapy with immunotherapies or other immune-enhancing modalities holds the potential for synergistic effects, leading to improved treatment outcomes (Jungles et al. 2022).

Immune System Supportive Care Measures during Chemotherapy

During chemotherapy, supportive care measures targeting the immune system can help mitigate immune suppression and enhance patients' overall well-being (Silk et al. 2022). Supportive care interventions may include optimizing nutrition, managing side effects promptly, and administering prophylactic treatments to prevent complications (Silk et al. 2022). Maintaining a healthy lifestyle can also support immune function during chemotherapy (Silk et al. 2022). Implementing these strategies to mitigate chemotherapy-induced immune suppression can enhance the effectiveness of cancer treatment and improve patient outcomes. Further research is needed to optimize these approaches' timing, sequencing, and combination, considering tumor-specific characteristics and individual patient factors.

Clinical Implications

Understanding chemotherapy's effects on the immune system has significant clinical implications for cancer treatment and patient management. Here are some key clinical implications:

Implications for Cancer Treatment and Patient Outcomes

Treatment efficacy and response

Chemotherapy-induced immune suppression can impact treatment efficacy and patient response. Immune cells, such as T-lymphocytes and natural killer cells, play critical roles in recognizing and eliminating cancer cells (Gonzalez et al. 2018). The immune suppression caused by chemotherapy may compromise the immune system's ability to mount an effective anti-tumor response, reducing treatment efficacy and potentially affecting patient outcomes. Therefore, considering the immune status of patients and implementing strategies to mitigate immune suppression can help optimize treatment responses.

Immune-related adverse events

Chemotherapy-induced immune suppression can increase the risk of immune-related adverse events. Immunosuppressed patients may be more susceptible to infections, opportunistic pathogens, and other immune-related toxicities. Managing and monitoring these adverse events are essential to ensure patient safety and treatment continuation (Topalian et al. 2012). Identifying patients at higher risk for immune-related complications and implementing preventive measures can minimize treatment interruptions and improve patient outcomes.

Predictive biomarkers and treatment selection

Understanding the impact of chemotherapy on the immune system can guide the identification of predictive biomarkers for treatment response and help in treatment selection (Bai et al. 2020). Biomarkers, such as immune cell profiles, cytokine signatures, or immune-related gene expression patterns, may help stratify patients who are more likely to benefit from specific chemotherapy regimens or combination therapies. Tailoring treatment approaches based on predictive biomarkers can improve treatment outcomes by targeting patients most likely to respond.

Combination therapies and immunomodulation

The recognition of chemotherapy's effect on the immune system has paved the way for combination therapies and immunomodulation strategies. Combining chemotherapy with immune checkpoint inhibitors, adoptive cell therapies, or other immunotherapies can synergistically enhance the anti-tumor immune response and improve treatment outcomes. Furthermore, immunomodulatory agents as adjuvants to chemotherapy can mitigate immune suppression and enhance treatment efficacy, potentially leading to better patient outcomes.

Long-term survival and immune function

Chemotherapy-induced immune suppression may affect long-term survival and immune function. Long-term immune effects, such as persistent immune dysregulation or immunosenescence, can impact long-term outcomes and patient quality of life. Understanding the long-term immune consequences of chemotherapy can help guide survivorship care and the development of strategies to optimize immune function and overall health in cancer survivors (Gennery et al. 2019). By comprehensively considering the implications of chemotherapy-induced immune suppression, clinicians can tailor treatment approaches, identify patients who may benefit from specific therapies, and implement strategies to mitigate immune-related complications. This personalized approach to cancer treatment can improve patient outcomes, enhance treatment efficacy, and optimize long-term survival and quality of life.

Potential Biomarkers to Predict Chemotherapy-Induced Immune Suppression

Identifying potential biomarkers to predict chemotherapy-induced immune suppression can aid in patient stratification, treatment selection, and the development of personalized treatment strategies. While research in this area is ongoing, several biomarkers show promise in predicting chemotherapy-induced immune suppression (Havel et al. 2019). Here are some potential biomarkers:

Lymphocyte subset analysis

Assessing the distribution and function of lymphocyte subsets, such as CD4+ T-cells, CD8+ T-cells, and regulatory T-cells (Tregs), can provide insights into immune suppression. Decreased CD4+ and CD8+ T-cell counts and an altered ratio of CD4+/CD8+ T-cells may indicate immune dysfunction and increased susceptibility to immune suppression (Bolivar-Wagers et al. 2022).

Cytokine and chemokine profiles

Monitoring cytokine and cytokine levels changes can provide valuable information about immune modulation. Decreased levels of pro-inflammatory cytokines, such as interferon-gamma (IFN-γ) or tumor necrosis factor-alpha (TNF-α), and increased levels of anti-inflammatory

cytokines, such as interleukin-10 (IL-10), may suggest immune suppression.

Immune checkpoint expression

Assessing the expression levels of immune checkpoint molecules, such as programmed cell death protein 1 (PD-1) and cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), on immune cells can indicate immune dysfunction. (Tu et al. 2020) Upregulation of immune checkpoint molecules may indicate immune exhaustion and compromised anti-tumor immune responses.

Immune-related gene expression patterns

Examining gene expression patterns associated with immune function and regulation can provide insights into chemotherapy-induced immune suppression. Genes involved in T-cell activation, antigen presentation, and immune cell signaling may show differential expression patterns in the context of immune suppression.

Peripheral blood immune cell profiling

Comprehensive profiling of immune cell subsets in peripheral blood, including myeloid-derived suppressor cells (MDSCs), natural killer (NK) cells, and dendritic cells (DCs), can offer valuable information about immune status (Wu et al. 2023). Alterations in these immune cell populations' frequencies or functional properties may reflect chemotherapy-induced immune suppression. It is important to note that these biomarkers' validation and clinical utility are still being explored, and their incorporation into routine clinical practice requires further research and validation studies. Additionally, considering the dynamic nature of the immune system and the heterogeneity of cancer types and chemotherapy regimens, a combination of multiple biomarkers and comprehensive immune profiling may yield more accurate predictions of chemotherapy-induced immune suppression.

Emerging Therapeutic Approaches for Minimizing Immune Suppression

Emerging therapeutic approaches aim to minimize immune suppression induced by chemotherapy, thereby enhancing the anti-tumor immune response and improving treatment outcomes. Here are some promising strategies currently being explored:

Targeted immunomodulation

Novel immunomodulatory agents are being developed to specifically target and modulate immune cells affected by chemotherapy-induced immune suppression. These agents can enhance immune cell activation, restore immune function, and overcome immune inhibitory signals. Examples include agonists of immune co-stimulatory receptors, such as OX40 or 4-1BB, which can enhance T-cell activation and effector functions. Other strategies involve targeting specific immune cell subsets or immune regulatory pathways to restore immune balance and promote anti-tumor immunity (Khalil et al. 2016).

Metabolic modulation

- Metabolic reprogramming of immune cells is an emerging area of research to mitigate immune suppression. Chemotherapy-induced immune suppression can alter the metabolic profiles of immune cells, leading to impaired function. Modulating metabolic pathways, such as enhancing glycolysis or fatty acid oxidation, can restore immune cell activity and promote anti-tumor immune responses (Sun et al. 2020). Metabolic regulators, including small molecules or metabolic modulators, are being investigated as potential therapeutic interventions to improve immune cell function and counteract immune suppression.

Microbiota modulation

The gut microbiota regulates immune responses and modulates treatment outcomes (Wei et al. 2018). Chemotherapy can disrupt the composition and diversity of the gut microbiota, affecting immune function. Modulating the microbiota through probiotics, prebiotics, or fecal microbiota transplantation (FMT) is an emerging approach to minimize immune suppression. Restoring a healthy and diverse microbiota may enhance immune cell activation and anti-tumor immunity, potentially improving treatment responses.

Nanoparticle-based drug delivery systems

Nanoparticle-based drug delivery systems offer opportunities to mitigate immune suppression by improving drug delivery to target tissues and minimizing off-target effects. Nanoparticles can be engineered to encapsulate chemotherapy agents and selectively deliver them to tumor sites, reducing systemic exposure and associated immune toxicity (Binnewies et al. 2018). Nanoparticles can also be designed to modulate immune responses by incorporating

immune-stimulating molecules or checkpoint inhibitors, enhancing the anti-tumor immune response while minimizing immune suppression.

Combination therapies with immunomodulators

Combining chemotherapy with immunomodulators, such as immune checkpoint inhibitors or cytokines, is an area of active investigation. The rationale behind combination therapies is to capitalize on the cytotoxic effects of chemotherapy while simultaneously enhancing immune activation and anti-tumor immune responses (Binnewies et al. 2018). These synergistic approaches aim to maximize the therapeutic benefits by simultaneously targeting cancer cells and boosting the immune system. Developing and implementing these emerging therapeutic approaches hold promise for minimizing immune suppression induced by chemotherapy. Continued research and clinical trials will provide further insights into their efficacy, safety, and potential synergies with existing treatment modalities.

Future direction

The future directions in chemotherapy-induced immune modulation encompass several key aspects (Table 3). Firstly, there is a need for further research to identify predictive biomarkers that can guide treatment decisions and stratify patients who are likely to benefit from specific immunomodulatory approaches in combination with chemotherapy. This would enable a more personalized and targeted approach to cancer treatment, optimizing therapeutic outcomes while minimizing unnecessary toxicities. Additionally, ongoing studies should focus on elucidating the mechanisms underlying chemotherapy-induced immune modulation to identify novel targets for intervention and develop more effective immunomodulatory strategies. Exploring the interplay between the tumor microenvironment, immune cells, and chemotherapy will shed light on the complex interactions and help design tailored approaches to enhance anti-tumor immune responses. Furthermore, clinical trials should investigate combination therapies' optimal timing, sequence, and dosing regimens to maximize their synergistic effects and minimize potential antagonisms. A comprehensive evaluation of treatment efficacy, long-term outcomes, and the impact on patient quality of life will provide valuable insights into the clinical benefits of these combined approaches. Overall, future research efforts should focus on refining and expanding the therapeutic potential of immunomodulatory agents in combination with chemotherapy, ultimately improving patient outcomes and advancing the field of cancer immunotherapy.

Conclusion

Chemotherapy-induced immune suppression is a complex phenomenon that can significantly impact cancer treatment outcomes. Understanding the mechanisms underlying immune suppression and its implications for cancer treatment and patient outcomes is crucial for optimizing therapeutic strategies. Chemotherapy targets cancer cells and affects immune cells, leading to compromised immune function, altered immune cell differentiation and function, and disrupted immune cell communication and signaling pathways. This immune suppression can hamper the anti-tumor immune response and increase the risk of immune-related toxicities. However, emerging therapeutic approaches show promise in mitigating chemotherapy-induced immune suppression. Strategies such as immunomodulatory agents as adjuvants to chemotherapy, combination therapies to enhance immune function, and immune system supportive care measures can help counteract immune suppression and improve treatment outcomes. Additionally, identifying potential biomarkers to predict immune suppression and guide treatment decisions holds great potential for personalized cancer medicine. Future directions in this field involve further elucidating the underlying mechanisms, exploring novel combination therapies, and developing targeted interventions to minimize immune suppression. The validation

Table 3 Potential Future Directions in Chemotherapy-Induced Immune Modulation

Aspect	Scope
Biomarker discovery	Identify predictive biomarkers for personalized treatment selection.
Mechanistic understanding	Elucidate the mechanisms underlying chemotherapy-induced immune modulation for target identification.
Tumor microenvironment exploration	Investigate the interplay between the tumor microenvironment, immune cells, and chemotherapy.
Optimal combination strategies	Determine the optimal timing, sequence, and dosing regimens of combination therapies.
Clinical outcomes assessment	Evaluate treatment efficacy, long-term outcomes, and impact on patient quality of life.

The table briefly overviews potential future directions in chemotherapy-induced immune modulation, highlighting key aspects and their scope in advancing the field.

and clinical utility of predictive biomarkers and the development of innovative approaches like targeted immunomodulation and nanoparticle-based drug delivery systems offer exciting research and clinical application avenues. In conclusion, we can optimize cancer treatment strategies by understanding and mitigating chemotherapy-induced immune suppression, enhancing the anti-tumor immune response, and improving patient outcomes. Continued research, collaboration, and translation of findings into clinical practice will contribute to advancing this field and ultimately benefit cancer patients worldwide.

Author Contributions Conceptualization, methodology, validation, writing—review, SJ., A.K and AS.; formal analysis, resources, AS, SJ., and A.K.; data curation, AS, and SJ. All authors have read and agreed to the published version of the manuscript. The authors confirm that no paper mill and artificial intelligence was used.

Funding This manuscript did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability This document includes citations for all the data that were analyzed throughout the literature review.

Declarations

Ethical approval and Clinical trials Not applicable

Consent to participate Not applicable

Consent for publication All authors have their consent to publish.

Competing interests The authors declare no competing interests.

REFERENCES

Abu-Eid R, Samara RN, Ozbun L, Abdalla MY, Berzofsky JA, Friedman KM, Mkrtichyan M (2017) Selective inhibition of regulatory T-cells by targeting the PI3K-Akt pathway. Cancer Immunology Research 5(7):563–575. https://doi.org/10.1158/2326-6066.CIR-16-0337

American Cancer Society. (2021). Chemotherapy. Retrieved from https://www.cancer.org/treatment/treatments-and-side-effects/treatment-types/chemotherapy.html

Ansell SM, Lesokhin AM, Borrello I, Halwani A, Scott EC, Gutierrez M et al (2015) PD-1 blockade with nivolumab in relapsed or refractory Hodgkin's lymphoma. New England Journal of Medicine 372(4):311–319. https://doi.org/10.1056/NEJMoa1411087

Bai R, Lv Z, Xu D, Cui J (2020) Predictive biomarkers for cancer immunotherapy with immune checkpoint inhibitors. Biomark Res 8:1–17. https://doi.org/10.1186/s40364-020-00209-0

Beghdadi W, Madjene LC, Benhamou M, Charles N, Gautier G, Launay P, Blank U (2011) MasT-cells as cellular sensors in inflammation and immunity. Frontiers in immunology 2:37. https://doi.org/10.3389/fimmu.2011.00037

Bende RJ, Aarts WM, van Noesel CJ (2011) Targeting the B-cell receptor pathway in diffuse large B-cell lymphoma. Leukemia 25(3):387–398. https://doi.org/10.1038/leu.2010.288

Binnewies M, Roberts EW, Kersten K, Chan V, Fearon DF, Merad M et al (2018) Understanding the tumor immune microenvironment (TIME) for effective therapy. Nature Medicine 24(5):541–550. https://doi.org/10.1038/s41591-018-0014-x

Bolivar-Wagers S, Larson JH, Jin S, Blazar BR (2022) Cytolytic CD4+ and CD8+ regulatory T-cells and implications for developing immunotherapies to combat graft-versus-host disease. Front Immunol 13:1–21. https://doi.org/10.3389/fimmu.2022.864748

Cao Y, Wang X, Jin T, Tian Y, Dai C, Widarma C et al (2020) Immune checkpoint molecules in natural killer cells as potential targets for cancer immunotherapy. Signal transduction and targeted therapy 5(1):250. https://doi.org/10.1038/s41392-020-00318-w

Caro JJ, Salas M, Ward A, Goss G (2001) Anemia as an independent prognostic factor for survival in patients with cancer: a systematic, quantitative review. Cancer 91(12):2214–2221. https://doi.org/10.1002/1097-0142(20010615)91:12<2214::aid-cncr1274>3.0.co;2-p

Casadevall A, Dadachova E (2004) Antibody-mediated chemotherapy: recent advances and future prospects. Trends in Biotechnology 22(9):424–430. https://doi.org/10.1016/j.tibtech.2004.06.010

Casares N, Pequignot MO, Tesniere A, Ghiringhelli F, Roux S, Chaput N et al (2005) Caspase-dependent immunogenicity of doxorubicin-induced tumor cell death. The Journal of Experimental Medicine 202(12):1691–1701. https://doi.org/10.1084/jem.20050915

Chao MP, Alizadeh AA, Tang C, Myklebust JH, Varghese B, Gill S et al (2010) Anti-CD47 antibody synergizes with rituximab to promote phagocytosis and eradicate non-Hodgkin lymphoma. Cell 142(5):699–713. https://doi.org/10.1016/j.cell.2010.07.044

Choudhury N, Nakka KK, Choudhury SR, Konar S (2018) Immune system modulation and role of cytokines as immunotherapy in cancer treatment. International Journal of Cell Biology 2018:5167342. https://doi.org/10.1155/2018/5167342

Constantino J, Gomes C, Falcão A, Neves BM, Cruz MT (2017) Dendritic cell-based immunotherapy: a basic review and recent advances. Immunologic research 65:798–810. https://doi.org/10.1007/s12026-017-8957-6

Coventry BJ (2019) Therapeutic vaccination immunomodulation: forming the basis of all cancer immunotherapy. Therapeutic Advances in Vaccines and Immunotherapy.:7. https://doi.org/10.1177/2515135519862234

Crotty S (2019) T follicular helper cell biology: a decade of discovery and diseases. Immunity 50(5):1132–1148. https://doi.org/10.1016/j.immuni.2019.04.011

De Milito A, Younes A (2008) The central role of lymphocytes in the cancer microenvironment. Oncology 72(3-4):124–132. https://doi.org/10.1159/000151563

Del Prete A, Salvi V, Soriani A et al (2023) Dendritic cell subsets in cancer immunity and tumor antigen sensing. *Cell Mol Immunol* **20**:432–447. https://doi.org/10.1038/s41423-023-00990-6

Deptuła M, Zieliński J, Wardowska A, Pikuła M (2019) Wound healing complications in oncological patients: perspectives for cellular therapy. Advances in Dermatology and Allergology/PostępyDermatologiiiAlergologii 36(2):139–146. https://doi.org/10.5114/ada.2019.84517

Emens LA (2018) Breast cancer immunotherapy: Facts and hopes. Clinical Cancer Research 24(3):511–520. https://doi.org/10.1158/1078-0432.CCR-16-3001

Facciabene A, Motz GT (2012) Tumor-infiltrating lymphocytes: A critical review of the literature. Journal of Immunotherapy 35(7):585–598. https://doi.org/10.1097/CJI.0b013e318275eebf

Fridman WH, Pagès F, Sautès-Fridman C, Galon J (2012) The immune contexture in human tumours: Impact on clinical outcome. Nature Reviews Cancer 12(4):298–306. https://doi.org/10.1038/nrc3245

Gajewski TF, Meng Y, Blank C, Brown I, Kacha A, Kline J, Harlin H (2006) Immune resistance orchestrated by the tumor microenvironment. Immunological Reviews 213(1):131–145. https://doi.org/10.1111/j.1600-065X.2006.00442.x

Gajewski TF, Schreiber H, Fu YX (2013) Innate and adaptive immune cells in the tumor microenvironment. Nature Immunology 14(10):1014–1022. https://doi.org/10.1038/ni.2703

Galluzzi L, Buqué A, Kepp O, Zitvogel L, Kroemer G (2015) Immunogenic cell death in cancer and infectious disease. Nature Reviews Immunology 17(2):97–111. https://doi.org/10.1038/nri.2016.107

Galluzzi L, Senovilla L, Zitvogel L, Kroemer G (2012) The secret ally: immunostimulation by anticancer drugs. Nature Reviews. Drug Discovery 11(3):215–233. https://doi.org/10.1038/nrd3626

Gao L, Zhang L, Hu J, Li F, Shang Z (2020a) Immune checkpoint inhibitors combination therapy for lymphoma: current knowledge and future directions. Frontiers in Immunology 11:1679. https://doi.org/10.3389/fimmu.2020.01679

Gao Y, Souza-Fonseca-Guimaraes F, Bald T, Smyth MJ (2020b) Endogenous tumor-reactive CD8+ T-cells in cancer immunotherapy: Potentials and challenges. Frontiers in Immunology 11:1–18. https://doi.org/10.3389/fimmu.2020.00786

Garg AD, Agostinis P (2017) Cell death and immunity in cancer: From danger signals to mimicry of pathogen defense responses. Immunological Reviews 280(1):126–148. https://doi.org/10.1111/imr.12574

Gennery AR, Lankester A, Inborn Errors Working Party (IEWP) of the European Society for Blood and Marrow Transplantation (EBMT) (2019) Long term outcome and immune function after hematopoietic stem cell transplantation for primary immunodeficiency. Front Pediatr 24(7):1–7. https://doi.org/10.3389/fped.2019.00381

Gonzalez H, Hagerling C, Werb Z (2018) Roles of the immune system in cancer: from tumor initiation to metastatic progression. Genes Dev 32(19–20):1267–1284. https://doi.org/10.1101/gad.314617.118

Havel JJ, Chowell D, Chan TA (2019) The evolving landscape of biomarkers for checkpoint inhibitor immunotherapy. Nat Rev Cancer 19(3):133–150. https://doi.org/10.1038/s41568-019-0116-x

Hurez V, Padron ÁS, Svatek RS, Curiel TJ, Kim J (2015) Immune-based therapy in metastatic renal cell carcinoma: current and future perspectives. Immunotherapy 7(2):191–202. https://doi.org/10.2217/imt.14.102

Janssens S, Beyaert R (2003) Role of Toll-like receptors in pathogen recognition. Clinical microbiology reviews 16(4):637–646. https://doi.org/10.1128/CMR.16.4.637-646.2003

Joseph GS, Joseph CH (2006) Evidence accumulating in support of cancer vaccines combined with chemotherapy: A pragmatic review of past and present efforts. International Journal of Oncology 29(4):765–777. https://doi.org/10.3892/ijo.29.4.765

Joshi SS, Badgwell BD (2021) Current treatment and recent progress in gastric cancer. CA: A Cancer Journal for Clinicians 71(3):264–279. https://doi.org/10.3322/caac.21627

Jungles KM, Holcomb EA, Pearson AN, Jungles KR, Bushop CR, Pierce LJ, Green MD, Speers CW (2022) Updates in combined approaches of radiotherapy and immune checkpoint inhibitors for the treatment of breast cancer. Front. Oncol. 12. https://doi.org/10.3389/fonc.2022.1022542

Katz J, Feldman MA, Bass EB, Lubomski LH, Tielsch JM, Petty BG et al (2001) Adverse intraoperative medical events and their association with anesthesia management strategies in cataract surgery. Ophthalmology 108(10):1721–1726. https://doi.org/10.1016/S0161-6420(01)00696-7

Khalil DN, Smith EL, Brentjens RJ, Wolchok JD (2016) The future of cancer treatment: immunomodulation, CARs and combination immunotherapy. Nat Rev Clin Oncol 13(5):273–290. https://doi.org/10.1038/nrclinonc.2016.25

Klapper JA, Downey SG, Smith FO, Yang JC, Hughes MS, Kammula US et al (2008) High-dose interleukin-2 for the treatment of metastatic renal cell carcinoma: a retrospective analysis of response and survival in patients treated in the surgery branch at the National Cancer Institute between 1986 and 2006. Cancer 113(2):293–301. https://doi.org/10.1002/cncr.23552

Koh AY, Priebe GP, Ray C, Van Rooijen N, Pier GB (2009) Inescapable need for neutrophils as mediators of cellular innate immunity to acute Pseudomonas aeruginosa pneumonia. Infection and immunity 77(12):5300–5310. https://doi.org/10.1128/IAI.00781-09

Kroschinsky F, Stölzel F, von Bonin S, Beutel G, Kochanek M, Kiehl M et al (2017) New drugs, new toxicities: severe side effects of modern targeted and immunotherapy of cancer and their management. Critical Care 21:1–11. https://doi.org/10.1186/s13054-016-1588-4

Lana S, U'ren L, Plaza S, Elmslie R, Gustafson D, Morley P, Dow S (2007) Continuous low-dose oral chemotherapy for adjuvant therapy of splenic hemangiosarcoma in dogs. Journal of Veterinary Internal Medicine 21(4):764–769. https://doi.org/10.1111/j.1939-1676.2007.tb03096.x

Lee L, Gupta M, Sahasranaman S (2016) Immune checkpoint inhibitors: An introduction to the next-generation cancer immunotherapy. The Journal of Clinical Pharmacology 56(2):157–169. https://doi.org/10.1002/jcph.596

Lee YT, Tan YJ, Oon CE (2018) Molecular targeted therapy: Treating cancer with specificity. European Journal of Pharmacology 834:188–196. https://doi.org/10.1016/j.ejphar.2018.07.010

Lenz G, Wright GW, Dave SS (2015) The role of the tumor microenvironment in chemotherapy-resistant lymphoma. Cancer Microenvironment 8(2):59–73. https://doi.org/10.1007/s12307-014-0159-3

Levi M, Salaroli R, Parenti F, Maria R D, Zannoni A, Bernardini C, Gola C, Brocco A, Marangio A, Benazzi C, Muscatello L V, Brunetti B, Forni M, Sarli G, National institute of health 2018 https://doi.org/10.1186/s12917-020-02709-5

Ma Y, Adjemian S, Mattarollo SR, Yamazaki T, Aymeric L, Yang H et al (2013) Anticancer chemotherapy-induced intratumoral recruitment and differentiation of antigen-presenting cells. Immunity 38(4):729–741. https://doi.org/10.1016/j.immuni.2013.03.003

Mackall CL, Fleisher TA, Brown MR, Magrath IT, Shad AT, Horowitz ME et al (1995) Lymphocyte depletion during treatment with intensive chemotherapy for cancer. Blood 84(7):2221–2228. https://doi.org/10.1182/blood.v84.7.2221.bloodjournal8472221

Merle NS, Noe R, Halbwachs-Mecarelli L, Fremeaux-Bacchi V, Roumenina LT (2015) Complement system part II: role in immunity. Frontiers in immunology 6:257. https://doi.org/10.3389/fimmu.2015.00257

Mokhtari RB, Homayouni TS, Baluch N, Morgatskaya E, Kumar S, Das B, Yeger H (2017) Combination therapy in combating cancer. Oncotarget 8(23):38022–38043. https://doi.org/10.18632/oncotarget.16723

Nimmerjahn F, Ravetch JV (2008) Fcγ receptors: old friends and new family members. Immunity 28(1):18–28. https://doi.org/10.1016/j.immuni.2007.11.020

O'Donnell JS, Teng MWL, Smyth MJ (2019) Cancer immunoediting and resistance to T cell-based immunotherapy. *Nat Rev Clin Oncol 16*:151–167. https://doi.org/10.1038/s41571-018-0142-8

Pitt JM, Marabelle A, Eggermont A, Soria JC, Kroemer G, Zitvogel L (2016) Targeting the tumor microenvironment: Removing obstruction to anticancer immune responses and immunotherapy. Annals of Oncology 27(8):1482–1492. https://doi.org/10.1093/annonc/mdw168

Pone EJ, Zan H (2012) Antibody isotype diversification and the regulation of antibody responses. The Journal of Immunology 189(12):5173–5174. https://doi.org/10.4049/jimmunol.1290061

Postow MA, Callahan MK, Wolchok JD (2015) Immune checkpoint blockade in cancer therapy. Journal of Clinical Oncology 33(17):1974–1982. https://doi.org/10.1200/JCO.2014.59.4358

Poulain S, Roumier C, Venet F, Lepape A, Meilleroux J (2019) Plasma cells: from production to pathologic disorders. European Journal

of Internal Medicine 62:10–16. https://doi.org/10.1016/j.ejim.2018.09.015

Rabinowich H, Cohen R, Bruderman I, Steiner Z, Klajman A, Fride E et al (1994) Regression of established murine carcinoma metastases following vaccination with tumor-associated antigen peptides. Cancer Research 54(4):1049–1057

Ramakrishnan R, Huang C, Cho HI (2020) Immunomodulatory effects of conventional chemotherapy and targeted therapy on the tumor microenvironment. Advances in Experimental Medicine and Biology 1248:83–96. https://doi.org/10.1007/978-3-030-38397-7_4

Reits EA, Hodge JW, Herberts CA, Groothuis TA, Chakraborty M, Wansley EK et al (2006) Radiation modulates the peptide repertoire, enhances MHC class I expression, and induces successful antitumor immunotherapy. The Journal of Experimental Medicine 203(5):1259–1271. https://doi.org/10.1084/jem.20052494

Restifo NP, Dudley ME, Rosenberg SA (2012) Adoptive immunotherapy for cancer: Harnessing the T-cell response. Nature Reviews Immunology 12(4):269–281. https://doi.org/10.1038/nri3191

Ruffell B, Coussens LM (2015) Macrophages and therapeutic resistance in cancer. Cancer Cell 27(4):462–472. https://doi.org/10.1016/j.ccell.2015.02.015

Schatton T, Schütte U, Frank NY, Zhan Q, Hoerning A, Robles SC, Zhou J, Hodi FS, Spagnoli GC, Murphy GF, Frank MH (2010) Modulation of T-cell activation by malignant melanoma initiating cells. Cancer Res 70(2):697–708. https://doi.org/10.1158/0008-5472.CAN-09-1592

Schirrmacher V (2019) From chemotherapy to biological therapy: A review of novel concepts to reduce the side effects of systemic cancer treatment (Review). International Journal of Oncology 54:407–419. https://doi.org/10.3892/ijo.2018.4661

Seo H, Jeong HW (2020) Chemotherapy-induced immunosuppression and tumor immune evasion. Immune Network 20(1):e9. https://doi.org/10.4110/in.2020.20.e9

Seruga B, Zhang H, Bernstein LJ, Tannock IF (2008) Cytokines and their relationship to the symptoms and outcome of cancer. Nature Reviews Cancer 8(11):887–899. https://doi.org/10.1038/nrc2501

Shalapour S, Karin M (2015) Immunity, inflammation, and cancer: an eternal fight between good and evil. The Journal of Clinical Investigation 125(9):3347–3355. https://doi.org/10.1172/JCI80007

Shapouri-Moghaddam A, Mohammadian S, Vazini H, Taghadosi M, Esmaeili SA, Mardani F et al (2018) Macrophage plasticity, polarization, and function in health and disease. Journal of cellular physiology 233(9):6425–6440. https://doi.org/10.1002/jcp.26429

Sharma P, Allison JP (2015) Immune checkpoint targeting in cancer therapy: Toward combination strategies with curative potential. Cell 161(2):205–214. https://doi.org/10.1016/j.cell.2015.03.030

Shin DS, Ribas A (2015) The evolution of checkpoint blockade as a cancer therapy: what's here, what's next? Current Opinion in Immunology 33:23–35. https://doi.org/10.1016/j.coi.2015.01.006

Silk AW, Barker CA, Bhatia S, Bollin KB, Chandra S, Eroglu Z, Gastman BR, Kendra KL, Kluger H, Lipson E.J. Madden, K, (2022) Society for Immunotherapy of Cancer (SITC) clinical practice guideline on immunotherapy for the treatment of nonmelanoma skin cancer. J Immunother Cancer 10(7):1–27. https://doi.org/10.1136/jitc-2021-004434

Sistigu A, Yamazaki T, Vacchelli E, Chaba K, Enot DP, Adam J et al (2014) Cancer cell–autonomous contribution of type I interferon signaling to the efficacy of chemotherapy. Nature Medicine 20(11):1301–1309. https://doi.org/10.1038/nm.3708

Spranger S, Gajewski TF (2018) Impact of oncogenic pathways on evasion of antitumor immune responses. Cancer Discovery 8(8):1069–1087. https://doi.org/10.1158/2159-8290.CD-18-0266

Sun L, Yang X, Yuan Z, Wang H (2020) Metabolic reprogramming in immune response and tissue inflammation. Arterioscler Thromb Vascul Biol 40(9):1990–2001. https://doi.org/10.1161/ATVBAHA.120.314037

Teoh F, Pavelka N (2016) How chemotherapy increases the risk of systemic candidiasis in cancer patients: current paradigm and future directions. Pathogens 5(1):6. https://doi.org/10.3390/pathogens5010006

Topalian SL, Drake CG, Pardoll DM (2015) Targeting the PD-1/B7-H1(PD-L1) pathway to activate anti-tumor immunity. Current Opinion in Immunology 27:1–7. https://doi.org/10.1016/j.coi.2014.11.001

Topalian SL, Hodi FS, Brahmer JR, Gettinger SN, Smith DC, McDermott DF et al (2012) Safety, activity, and immune correlates of anti-PD-1 antibody in cancer. New England Journal of Medicine 366(26):2443–2454. https://doi.org/10.1056/NEJMoa1200690

Tu L, Guan R, Yang H, Zhou Y, Hong W, Ma L, Zhao G, Yu M (2020) Assessment of the expression of the immune checkpoint molecules PD-1, CTLA4, TIM-3 and LAG-3 across different cancers in relation to treatment response, tumor-infiltrating immune cells and survival. Int J Cancer 147(2):423–39. https://doi.org/10.1002/ijc.32785

Tumeh PC, Harview CL, Yearley JH, Shintaku IP, Taylor EJM, Robert L et al (2014) PD-1 blockade induces responses by inhibiting adaptive immune resistance. Nature 515(7528):568–571. https://doi.org/10.1038/nature13954

Ujjani C, Wang H, Skarbnik A, Jha R (2020) Clinical updates in managing and monitoring adverse events from CAR T-cells. Current Hematologic Malignancy Reports 15(3):193–203. https://doi.org/10.1007/s11899-020-00579-y

Vincent J, Mignot G (2018) Immunotherapy: New Weapons for Cancer Treatment. Medecine Sciences 34(1):47–54. https://doi.org/10.1051/medsci/20183401010

Wang M, Zhao J, Zhang L, Wei F, Lian Y, Wu Y, Gong Z, Zhang S, Zhou J, Cao K, Li X, Xiong W, Li G, Zeng Z, Guo C (2017) Role of tumor microenvironment in tumorigenesis. J Cancer 8(5):761–773. https://doi.org/10.7150/jca.17648

Wei SC, Duffy CR, Allison JP (2018) Fundamental mechanisms of immune checkpoint blockade therapy. Cancer Discovery 8(9):1069–1086. https://doi.org/10.1158/2159-8290.CD-18-0367

Weiskopf K, Weissman IL (2021) Macrophages are critical effectors of antibody therapies for cancer. MAbs 13(1):1900202. https://doi.org/10.1080/19420862.2020.1900202

Wu B, Shi X, Jiang M, Liu H (2023) Cross-talk between cancer stem cells and immune cells: potential therapeutic targets in the tumor immune microenvironment. Mol Cancer 22(1):1–22. https://doi.org/10.1186/s12943-023-01748-4

Zitvogel L, Apetoh L, Ghiringhelli F, André F, Tesniere A, Kroemer G (2008) The anticancer immune response: indispensable for therapeutic success? Journal of Clinical Investigation 118(6):1991–2001

Zou W, Wolchok JD, Chen L (2016) PD-L1 (B7-H1) and PD-1 pathway blockade for cancer therapy: Mechanisms, response biomarkers, and combinations. Science Translational Medicine 8(328):328rv4. https://doi.org/10.1126/scitranslmed.aad7118

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.